# Winfried van den Dool

Amsterdam, The Netherlands
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## Education

2021 - present

**Ph.D. in Machine Learning**
QUVA Lab, University of Amsterdam, the Netherlands
Topic: Computational efficiency and hardware-aware methods in deep learning, (vector-)quantization, hierarchical architectures, adaptive compute allocation, inference-time optimization
Supervisor: Prof. dr. Max Welling

2016 - 2019

**Master Mathematical Sciences *(cum laude)*, Utrecht University**
Specialization in probability, statistics and mathematical modeling
Thesis: A new method using Bayesian Neural Networks to generate explanations of Deep Learning decisions (*grade 9/10*)
Final GPA: 4.00

2016 - 2017

**Premaster Economics, Utrecht University**
Final GPA: 4.00

2012 - 2016

**Bachelor Mathematics (*cum laude*), Utrecht University**
**Bachelor Physics and Astronomy (*cum laude*), Utrecht University**
Minors in: Computer Science (*grade 9/10*), Econometrics (*grade 9/10*) and Philosophy of Science (honors program)
Thesis: A new algorithm for counting self-avoiding walks: the Tiling method (*9/10*)
Final GPA: 4.00

## Work experience

2025

**Research Internship (6 months) at Fermioniq** on quantum algorithms for efficiently solving Partial Differential Equations using tensor network representations

2018 - 2021

**Data Scientist at PricewaterhouseCoopers** software developer focusing on the application of machine learning in finance, with an emphasis on explainable AI.

2018

**Research Internship (6 months) at RaboBank** doing causal inference research on macroeconomic variables and the influence of Quantitative Easing on yield curves.

## Publications

2026

**AVQ-Attention-2: Adaptive Vector-Quantized Attention with Exact Local Blocks and Linear-Cost Training** (under review)
Winfried van den Dool, Patrick Forré, Amir Habibian, Yuki M. Asano, Max Welling

2026

**AVQ-Attention: Adaptive Vector-Quantized Attention** (ECCV)
Winfried van den Dool, Patrick Forré, Amir Habibian, Yuki M. Asano, Max Welling

2025

**Adaptive Mesh Quantization for Neural PDE Solvers** (TMLR, ICLR J2C certificate)
Winfried van den Dool, Maksim Zhdanov, Yuki M. Asano, Max Welling

2023

**Efficient Neural PDE-Solvers using Quantization Aware Training** (ICCV workshop)
Winfried van den Dool, Tijmen Blankevoort, Yuki M. Asano, Max Welling